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RESEARCH ARTICLE

# Airport Capacity Prediction and Optimization Based on Semi-Varying Coefficient and Dynamic Programming

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**ABSTRACT** Air traffic congestion has emerged as a critical challenge in modern air traffic management. Adverse weather conditions and other factors frequently lead to reduced airport arrival capacity, resulting in flight delays. Accurate capacity prediction and optimization are essential for maximizing airport utilization and minimizing flight delays. This study introduces a capacity prediction model based on the semi-varying coefficient approach, utilizing real-world data from Guangzhou Baiyun Airport collected over a specific time period. Through 5-fold cross-validation, the semi-varying coefficient model is compared to the varying-coefficient model, demonstrating superior out-of-sample prediction capabilities. Subsequently, we establish and solve a flight arrival and departure flow allocation optimization model using dynamic programming to enhance traffic flow coordination. Analysis of case study results demonstrates that this method can generate optimal strategies for allocating arrival and departure traffic flows within airport capacity constraints, thereby minimizing overall flight delay costs.

**INDEX TERMS** Air traffic flow management, airport capability utilization, capacity prediction, flow allocation.

## I. INTRODUCTION

Imbalances between demand and capacity at various elements of the air transportation system are a major cause of flight delays worldwide. In order to deal with demand-capacity mismatches and mitigate the impacts of delays, air traffic flow management (ATFM) strategies are implemented to regulate the flows at a regional/national level. In order to determine the need for ATFM and plan the traffic regulation, traffic managers need to estimate throughput performance and set target flow rates at airport and airspace resources (see, e.g., [1]). Weather events significantly impact airport performance, often leading to operational delays when capacity is constrained. This study defines airport capacity as the sustainable hourly throughput of runways during high-demand periods, encompassing both the Airport Departure Rate (ADR) and Airport Arrival Rate (AAR). The evolution of probabilistic traffic flow management necessitates the integration of weather information into the system, providing effective capacity distribution and probability estimates as key decision-making criteria. Therefore, the capacity assessment of weather impact has become a key problem in air traffic flow management (see, e.g., [1]). Moreover, the rapid growth of air transport has exacerbated the imbalance between capacity and demand during adverse weather conditions, significantly contributing to flight delays (see, e.g., [2]). In order to provide reasonable flow control decisions and effective post operation analysis, it is necessary to establish a predicting model based on the relationship between weather conditions and airport capacity so as to achieve the purpose of forecasting the airport capacity. This study aims to address this need by developing a novel

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approach to airport capacity prediction and flow allocation optimization.

This paper proposes a capacity prediction model based on the semi-varying coefficient approach. Through comparison with the variable parameter model, we demonstrate that the semi-parametric model exhibits superior out-of-sample predictive capability. Subsequently, we develop an optimization model for arrival and departure flow allocation, constrained by airport capacity, with the objective of minimizing overall flight delay costs. We employ dynamic programming to solve this optimization model. Analysis of the computational results demonstrates that our approach can generate optimal strategies for arrival and departure flow allocation within airport capacity constraints, providing effective decision support for tactical air traffic flow management.

The remainder of this paper is structured as follows: Section II provides a comprehensive review of the existing literature relevant to the study. Section III presents the theoretical foundations and preliminaries of the models and algorithms used in this study. The first part of Section IV focuses on the development of prediction models for Airport Arrival Rate (AAR) and Airport Departure Rate (ADR) using a semi-varying coefficient model and real-world datasets. The second part of Section IV addresses the optimization and coordination of arrival and departure flow allocation for a specific time period, utilizing real airport data. Finally, Section V summarizes the key findings, discusses the implications of our research, and suggests directions for future work.

## II. LITERATURE REVIEW

### A. AIRPORT CAPACITY PREDICTION

Accurate airport capacity estimation is crucial for the secure and orderly operation of the aviation system. However, such estimation is a non-trivial task as capacity depends on various meteorological and operational features. The complex coupling characteristics among these multisource features have proved to be challenging for most of the traditional regression models. Wang et al. [3] proposed to use linear regression and quadratic response surface regression algorithms for airport capacity prediction at Chicago O'Hare International Airport. Cox and Kochenderfer [4] proposed a Bayesian regression-based algorithm and also provided insight into the importance of factors. Further, more comprehensive features were considered in airport capacity prediction. Tien et al. [5] considered runway configuration selection and the more complicated weather information obtained from the short range ensemble forecast (SREF). They validated the model by applying it to 35 major airports in the U.S. Murça and Hansman [6] included the flight trajectory data in the terminal area and used the Gauss regression algorithm to predict airport capacity and showed better performance compared with the former two algorithms. Their research was expanded by combining the cluster and regression algorithms [7]. However, due to the imbalance and diversity of features scatter, as the number of covariates increases, the estimation and prediction efficiency of nonparametric models are not ideal due to the curse of dimensionality. The 'curse of dimensionality' refers to the significant decline in estimation and prediction efficiency of nonparametric models as the number of covariates increases. This occurs because nonparametric models require an increasingly large amount of data to accurately estimate complex functional forms in high-dimensional spaces. Empirical studies show that applications involving low signal-noise ratio (SNR) and a large number of explanatory variables, the prediction performance of multi-dimensional nonparametric models is often unsatisfactory (see, e.g., [8], [9]), which often leads to the problem of dimensional disaster, so it is necessary to consider the semi parametric model. Nonparametric models are advantageous due to their flexibility and minimal assumptions about data distribution. However, this flexibility becomes a limitation in high-dimensional settings, where more data is needed to support the model's complexity, leading to potential inefficiencies. In contrast, the semi-parametric model mitigates the curse of dimensionality by introducing parameterization to reduce the degrees of freedom of nonparametric components. This enables the model to maintain high predictive performance even in high-dimensional data scenarios. On the one hand, the parameter part of the semi-parametric model still uses more sample information and has strong explanatory ability; On the other hand, it also relaxes some assumptions on explanatory variables, which makes the model more applicable and improves the accuracy of the model (see, e.g., [10], [11]). Semi-parametric models encompass various types, including partial linear models, varying coefficient models, and additive models, among others. In many cases, the prediction effect of semi-parametric model is better than that of ordinary multivariate linear model (see, e.g., [12]).

Obviously, the above methods did not consider the possible nonlinear relationship between response variables and covariates, as well as the interactive relationship between covariates (two covariates jointly affect the response variable). Therefore, this paper proposes a semi-variable coefficient model that can accommodate these relationships to study airport capacity prediction problems. The semi-varying coefficient model, as an important class of semiparametric models that has rapidly developed in recent years, is capable of not only reflecting both linear and nonlinear relationships between variables but also capturing the interaction effects among variables. On the one hand, airport capacity is influenced by a variety of complex factors, such as meteorological conditions (weather type, wind speed, temperature, etc.), runway configuration, flight traffic, and air traffic management strategies. These factors often have nonlinear relationships with airport capacity. Simple linear models are unable to capture these complex nonlinear relationships [13], whereas semi-varying coefficient models, by allowing certain coefficients to vary with covariates, can

better fit these nonlinear relationships [14]. On the other hand, airport capacity is not only influenced by individual covariates but also by the interactions between multiple covariates. For example, the combination of wind speed and runway direction can significantly impact the airport's throughput. Semi-varying coefficient models, by introducing interaction terms between covariates, can effectively capture these complex interaction effects [15]. The Semi-varying coefficient model has been widely studied in many fields, and it shows excellent performance in statistical analysis, see [16] and [17].

### B. FLOW ALLOCATION OPTIMIZATION

Following airport capacity prediction, it is essential to coordinate traffic flow allocation and develop air traffic flow management strategies to minimize flight delays. Flight arrival and departure flow allocation is a crucial air traffic flow management strategy that coordinates flow distribution based on air traffic demand and airport capacity, optimizing capacity utilization to minimize flight delays. Traditional air traffic strategic flow management regards arrival and departure as two independent processes, and airport capacity as a constant value. However, in reality, arrival and departure processes are interdependent and mutually constraining. When the traffic demand exceeds the airport capacity, the Pareto optimal point should be found on or below the capacity curve to minimize flight delay. Chen et al. [18] presents a novel computationally efficient algorithm to address uncertainty in air traffic flow management by using a chance-constrained optimization method. Gilbo [19] proposed a traffic assignment model to minimize the number of delayed flights in the arrival and departure queues. Jiang et al. [20] studied the traffic assignment problem of terminal area under the constraints of location and runway capacity, Dell'Olmo and Lulli [21] proposed a dynamic programming model of flow allocation established with the objective of minimizing flight delay. Ruan et al. [22] considered the influence of air traffic on the capacity of runway and corridor based on the idea of CDM (collaborative decision making), Chen and Deng [23], [24] established a multi-objective optimization model with the objective of capacity utilization satisfaction and airport system flow maximization.

However, there is still limited research on optimizing the delay cost, which is of great concern to airlines. This article takes minimizing delay cost as the optimization objective of the model, and uses a delay cost function that increases superlinear with delay time to describe the relationship between delay cost and delay time.

## III. THEORETICAL FOUNDATION AND METHODOLOGICAL OVERVIEW

This section provides a theoretical foundation for the key components of our study: semi-varying coefficient models, an improved profile least-square estimation method, and the flow allocation optimization model.

### A. SEMI-VARYING COEFFICIENT MODELS

The varying coefficient model is a versatile regression approach with broad applications across numerous fields. This model enhances flexibility and adaptability by treating the regression coefficients as unknown functions of other independent variables, rather than as constants. Because the system function is usually a univariate function of an independent variable, the problem of dimension disaster in fitting is effectively avoided. Let $Y$ as dependent variable, $X_1, X_2, \ldots, X_p$ and U are independent variables, $\alpha$, $\beta$ are parameters. If $Y, X_1, X_2, \ldots, X_p$ and U satisfy

$$Y = \alpha + \beta_1(U)X_1 + \beta_2(U)X_2 + \ldots + \beta_p(U)X_p + \epsilon, \tag{III.1}$$

model (III.1) called varying coefficient regression model. In the varying coefficient regression model (III.1), the coefficients are all functions of U. However, in practical problems, it is often encountered that some coefficients change with U, while the other parts are constants. Therefore, semi-varying coefficient models with more extensive application background can be obtained by properly adjusting the order of each item in the coefficient functions

$$\begin{aligned} Y = {} & \alpha + \beta_1(U)X_1 + \ldots + \beta_q(U)X_q + \beta_{q+1}X_{q+1} \\ & + \ldots + \beta_p X_p + \epsilon, \end{aligned} \tag{III.2}$$

with $\epsilon$ being the random error satisfying $E(\epsilon|U, X_1, \ldots, X_p) = 0$, $Var(\epsilon|U, X_1, \ldots, X_p) = \sigma^2$.

Let $(Y_i; X_{i1}, X_{i2}, \ldots, X_{ip}, U_i)$(i=1,2,...,n) is n groups of independent observation data, the sample form of semi-varying coefficient model is

$$\begin{aligned} Y_i = {} & \alpha + \beta_1(U_i)X_{i1} + \ldots + \beta_q(U_i)X_{iq} \\ & + \beta_{q+1}X_{i,q+1} + \ldots + \beta_p X_{i,p} + \epsilon_i. \end{aligned} \tag{III.3}$$

Compared with the varying coefficient models, the semi-varying coefficient models reflect the elaborate structure of the coefficient regression functions, which provide more detail information for analyzing and understanding the influence of independent variable on dependent variables. In practical applications, the distinction between varying and constant coefficients for observed data can be determined through statistical tests of coefficient significance. That is to say, under a certain significant level, if some coefficients do not change significantly, then these coefficients can be considered as constant coefficients.

The research results corresponding to Model (III.2) are relatively abundant For example, Zhou and You [25] estimated constant and variable coefficient functions using least squares and wavelet methods; Zhang et al. [26] proposed a two-stage estimation method for the coefficient function of a semi variable coefficient model based on local polynomial fitting, and provided the asymptotic bias and variance of the estimator. Fan and Huang [27] proposed a contour least squares estimation method for the coefficient function of a semi variable coefficient model based on local linear fitting, and proved the asymptotic normality of constant

coefficient estimation, further studying related statistical inference problems.

In this paper, on the basis of the least square estimation method based on local linear fitting for semi-varying coefficient model proposed by Fan and Huang [27], the local linear fitting method for varying coefficient functions is improved to local nonlinear fitting. The improved least square estimation for semi-varying coefficient model is obtained and it does not affect the asymptotic normality of constant coefficient.

### B. PROFILE LEAST-SQUARE ESTIMATION

This section introduce the profile least-square estimation method. In fact, the improved profile least square estimation method of semi-varying coefficient models is divided into two stages to estimate the coefficient function. Assuming that the constant coefficient is known, the semi-varying coefficient model is rewritten as the varying coefficient model, and the initial estimation of the varying coefficient function is obtained by using local nonlinear fitting. Then, the approximate linear regression model is obtained by substituting the original semi-varying coefficient model, and the constant coefficient estimation is obtained by using the usual least square estimation method, Finally, the constant coefficient is substituted back into the initial estimation of the varying coefficient function, and the improved estimation of the varying coefficient function is obtained.

The profile least-square estimation method for semi-varying coefficient models involves an iterative two-stage procedure to estimate both varying and constant coefficients. Below, we present the algorithm in a step-by-step manner to enhance clarity.

**Stage 1: Initial Estimation of Varying Coefficients**

#### 1) MODEL TRANSFORMATION

Assume the constant coefficients $\beta_{q+1}, \ldots, \beta_p$ are known. Rewrite the semi-varying coefficient model (III.3) as a standard varying coefficient model:

$$\begin{aligned} Z_i &= Y_i - \beta_{q+1}X_{i,q+1} - \ldots - \beta_p X_{i,p} \\ &= \alpha + \beta_1(U_i)X_{i1} + \ldots + \beta_q(U_i)X_{iq} + \epsilon_i. \end{aligned} \tag{III.4}$$

where $Z_i$ is the adjusted dependent variable.

#### 2) LOCAL NONLINEAR FITTING

For each $u_0 \in \mathscr{U}$, approximate the varying coefficient functions $\beta_j(u)$ using a quadratic Taylor expansion:

$$\begin{aligned} \beta_j(u) \approx{}& \beta_j(u_0) + \beta_j'(u_0)(u-u_0) \\ &+ \frac{\beta_j''(u_0)}{2}(u-u_0)^2, \quad j = 1, 2, \ldots, q. \end{aligned} \tag{III.5}$$

The local nonlinear fitting minimizes the following kernel-weighted least squares problem:

$$\sum_{i=1}^{n}\left[Z_i - \sum_{j=1}^{q}\left(\beta_j(u_0) + \beta_j'(u_0)(U_i-u_0) + \frac{\beta_j''(u_0)(U_i-u_0)^2}{2}\right)X_{ij}\right]^2 K_h(U_i-u_0) = \min. \tag{III.6}$$

where $K_h(t) = \frac{K(\frac{t}{h})}{h}$ is a kernel function with bandwidth $h$

#### 3) MATRIX REPRESENTATION

Define the design matrices $A_1, A_2, A_3$ and the kernel weight matrix $W(u_0)$:

$$A_1 = \begin{pmatrix} X_{11} & \cdots & X_{1q} \\ \vdots & \ddots & \vdots \\ X_{n1} & \cdots & X_{nq} \end{pmatrix}, \tag{III.7}$$

$$A_2 = A_1 \cdot \mathrm{diag}(U-u_0) \tag{III.8}$$

$$A_3 = A_1 \cdot \mathrm{diag}\left(\frac{(U-u_0)^2}{2}\right) \tag{III.9}$$

$$W(u_0) = \mathrm{Diag}\left(K_h(U_1-u_0), \ldots, K_h(U_n-u_0)\right). \tag{III.10}$$

The combined design matrix is $X_v(u_0) = [A_1 \quad A_2 \quad A_3]$, and the solution is:

$$\hat{a}(u_0) = \left(X_v^T(u_0)W(u_0)X_v(u_0)\right)^{-1} X_v^T(u_0)W(u_0)Z, \tag{III.11}$$

where $\hat{a}(u_0) = [\hat{\beta}_1(u_0), \ldots, \hat{\beta}_q(u_0), \hat{\beta}_1'(u_0), \ldots, \hat{\beta}_q''(u_0)]^T$.

**Stage 2: Estimation of Constant Coefficients**

#### 1) LINEAR MODEL SUBSTITUTION

Substitute the estimated varying coefficients $\hat{\beta}_j(U_i)$ back into the original semi-varying coefficient model (III.3) to form a linear regression model:

$$Y_i = \sum_{j=1}^{q}\hat{\beta}_j(U_i)X_{ij} + \sum_{j=q+1}^{p}\beta_j X_{ij} + \epsilon_i. \tag{III.12}$$

#### 2) ORDINARY LEAST SQUARES (OLS

we have

$$\begin{aligned} &\hat{\boldsymbol{\beta}}(u_0) = (\boldsymbol{I}_q, \boldsymbol{O}_q, \boldsymbol{O}_q)(\boldsymbol{X_v}^T(u_0)\boldsymbol{W}(u_0)\boldsymbol{X_v}(u_0))^{-1} \\ &\boldsymbol{X_v}^T(u_0)\boldsymbol{W}(u_0)\boldsymbol{Z}. \end{aligned} \tag{III.13}$$

$$\begin{aligned} &\hat{\boldsymbol{\beta}}'(u_0) = (\boldsymbol{O}_q, \boldsymbol{I}_q, \boldsymbol{O}_q)(\boldsymbol{X_v}^T(u_0)\boldsymbol{W}(u_0)\boldsymbol{X_v}(u_0))^{-1} \\ &\boldsymbol{X_v}^T(u_0)\boldsymbol{W}(u_0)\boldsymbol{Z}. \end{aligned} \tag{III.14}$$

$$\begin{aligned} &\hat{\boldsymbol{\beta}}''(u_0) = (\boldsymbol{O}_q, \boldsymbol{O}_q, \boldsymbol{I}_q)(\boldsymbol{X_v}^T(u_0)\boldsymbol{W}(u_0)\boldsymbol{X_v}(u_0))^{-1} \\ &\boldsymbol{X_v}^T(u_0)\boldsymbol{W}(u_0)\boldsymbol{Z}. \end{aligned} \tag{III.15}$$

where $\boldsymbol{I}_q, \boldsymbol{O}_q$ denotes p-order identity matrix and q-order zero matrix. Let

$$\boldsymbol{X_c} = \begin{pmatrix} X_{1,q+1} & X_{1,q+2} & \cdots & X_{1,p} \\ X_{2,q+1} & X_{2,q+2} & \cdots & X_{2,p} \\ \vdots & \vdots & \vdots & \vdots \\ X_{n,q+1} & X_{n,q+2} & \cdots & X_{n,p} \end{pmatrix},$$

$$\beta(c) = (\beta_{q+1}, \beta_{q+2}, \ldots, \beta_p)^T,$$

then $\boldsymbol{Z} = \boldsymbol{Y} - \boldsymbol{X}_c\boldsymbol{\beta}_c$, where $\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_n)^T$.

Denote $e_{j,3q}$ is the 3q-dimensional column vector with the j-th element is equal to 1 and the other elements are equal to 0, that

$$\hat{\boldsymbol{\beta}}_j(u_0) = e_{j,3q}^T\hat{\boldsymbol{a}}(u_0) = e_{j,3q}^T(\boldsymbol{X_v}^T(u_0)\boldsymbol{W}(u_0)\boldsymbol{X_v}(u_0))^{-1} \boldsymbol{X_v}^T(u_0)\boldsymbol{W}(u_0)\boldsymbol{Z}$$

Let $u_0 = U_1, U_2, \ldots, U_n$ respectively, we have

$$\hat{\boldsymbol{\beta}}_j(U_i) = e_{j,3q}^T\hat{\boldsymbol{a}}(U_i) = e_{j,3q}^T\left(\boldsymbol{X_v}^T(U_i)\boldsymbol{W}(U_i)\boldsymbol{X_v}(U_i)\right)^{-1} \boldsymbol{X_v}^T(U_i)\boldsymbol{W}(U_i)\left(\boldsymbol{Y} - \boldsymbol{X}_c\boldsymbol{\beta}_c\right)$$

Let

$$\boldsymbol{S}_v = \begin{pmatrix} (\boldsymbol{X}_1^T, \boldsymbol{O}_q, \boldsymbol{O}_q)(\boldsymbol{X}_v^T(U_1)\boldsymbol{W}(U_1)\boldsymbol{X}_v(U_1))^{-1}\boldsymbol{X}_v^T(U_1)\boldsymbol{W}(U_1) \\ (\boldsymbol{X}_2^T, \boldsymbol{O}_q, \boldsymbol{O}_q)(\boldsymbol{X}_v^T(U_2)\boldsymbol{W}(U_2)\boldsymbol{X}_v(U_2))^{-1}\boldsymbol{X}_v^T(U_2)\boldsymbol{W}(U_2) \\ \vdots \\ (\boldsymbol{X}_n^T, \boldsymbol{O}_q, \boldsymbol{O}_q)(\boldsymbol{X}_v^T(U_n)\boldsymbol{W}(U_n)\boldsymbol{X}_v(U_n))^{-1}\boldsymbol{X}_v^T(U_n)\boldsymbol{W}(U_n) \end{pmatrix}$$

take $\hat{\boldsymbol{\beta}}_j(U_i)$ into model (III.4), we get approximate model $\boldsymbol{Y} - \boldsymbol{X}_c\boldsymbol{\beta}_c = \boldsymbol{S}_v(\boldsymbol{Y} - \boldsymbol{X}_c\boldsymbol{\beta}_c) + \boldsymbol{\epsilon}$. Arranging the above formula, we have the linear regression model $(\boldsymbol{I} - \boldsymbol{S}_v)\boldsymbol{Y} = (\boldsymbol{I} - \boldsymbol{S}_v)\boldsymbol{X}_c\boldsymbol{\beta}_c) + \boldsymbol{\epsilon}$, then the estimation of constant coefficient vector and varying coefficient vector using the least squares estimation method [28] are

$$\hat{\boldsymbol{\beta}}_c = \left(\hat{\boldsymbol{\beta}}_{q+1}, \hat{\boldsymbol{\beta}}_{q+2}, \ldots, \hat{\boldsymbol{\beta}}_p\right)^T = \left(\boldsymbol{X}_c^T(\boldsymbol{I} - \boldsymbol{S}_v)^T(\boldsymbol{I} - \boldsymbol{S}_v)\boldsymbol{X}_c\right)^{-1}\boldsymbol{X}_c^T(\boldsymbol{I} - \boldsymbol{S}_v)^T(\boldsymbol{I} - \boldsymbol{S}_v)\boldsymbol{Y}.$$

$$\tilde{\boldsymbol{\beta}}(u_0) = (\boldsymbol{I}_q, \boldsymbol{O}_q, \boldsymbol{O}_q) \times \left(\boldsymbol{X}_v^T(u_0)\boldsymbol{W}(u_0)\boldsymbol{X}_v(u_0)\right)^{-1} \times \boldsymbol{X}_v^T(u_0)\boldsymbol{W}(u_0)\left(\boldsymbol{Y} - \boldsymbol{X}_c\hat{\boldsymbol{\beta}}_c\right).$$

Thus, the fitting value of dependent variable are

$$\hat{Y} = (\hat{Y}_1, \hat{Y}_2, \ldots, \hat{Y}_n)^T = \boldsymbol{S}_v(\boldsymbol{Y} - \boldsymbol{X}_c\hat{\boldsymbol{\beta}}_c) + \boldsymbol{X}_c\hat{\boldsymbol{\beta}}_c.$$

#### 3) ITERATIVE REFINEMENT

Repeat Stage 1 and Stage 2 until convergence. The stopping criterion is defined as:

$$\|\hat{\beta}_c^{(k)} - \hat{\beta}_c^{(k-1)}\| < 10^{-4}, \tag{III.16}$$

where $k$ denotes the iteration number

### C. FLOW ALLOCATION OPTIMIZATION MODEL

#### 1) MODEL BUILDING

This section of the paper employs optimization methods to prescribe Airport Arrival Rates (AAR) and Airport Departure Rates (ADR) for traffic flow management, utilizing the capacity predictions generated by our semi-varying coefficient model. In this model, we represent airport capacity as an envelope curve of arrival and departure capacity, rather than a single point. This approach captures the tradeoff between arrival and departure operations, which are defined as the number of flights taking off and landing at the airport within a specified time slot. As illustrated in Figure 1, point P represents the demand for arrivals and departures. When the demand point lies outside the area enclosed by the envelope curve, it indicates that the demand for arrivals and departures cannot be simultaneously met, signifying that the flow exceeds the capacity limit [29], [30]. In such cases, a decision must be made to allocate separate flows for arrivals and departures to satisfy the airport's capacity constraints. Periods requiring capacity allocation and flow distribution can be subdivided into several smaller decision intervals for more granular management. To implement this strategy effectively, we make the following assumptions:

1)The airport capacity envelope and arrival/departure demand are known for each decision period.

2)Flight delay time is defined as the product of the number of decision periods delayed and the length of each decision period. The delay cost for both arrivals and departures is represented by an identical exponential power function of the delay time. Individual variations in delay costs due to factors such as aircraft type and priority are not differentiated between arrival and departure flights.

3)We introduce a virtual time period following the last decision period, assuming infinite capacity during this interval. This assumption ensures that all flights eventually arrive or depart, enabling the calculation of total delay costs and ensuring the problem's solvability.

Our objective is to minimize the total delay cost of arrivals and departures throughout the entire duration while maximizing airport capacity utilization. To address this problem, we develop a dynamic programming model.

Parameters:

$\boldsymbol{T_n}$ : The decision time period is a set of sequential numbers $1, 2, \ldots, N, N+1$, where the $N+1$ period is a virtual period with unlimited capacity and a time duration of $\Delta T$.

$\boldsymbol{I_n}$: The n-th decision period capacity envelope boundary line segment number Collection;

$\hat{\boldsymbol{a_n}}, \hat{\boldsymbol{b_n}}$: The demand for arrivals and departures in the n-th time period, excluding the demand for delayed flights.

$\boldsymbol{\delta}$:the coefficient of the super-linear delay cost function

$\boldsymbol{\alpha_{i,n}}, \boldsymbol{\beta_{i,n}}, \boldsymbol{\gamma_{i,n}}$: the three-point function parameters of the i-th boundary line segment in the capacity envelope of the n-th decision period.

$\boldsymbol{a_n}, \boldsymbol{b_n}$:The number of delayed arrivals and departures from the n-th decision time period to the next time period.

$\boldsymbol{S_n\mathcal{D}(a_n, b_n)}$: representing the state of the n-th decision period.

$\boldsymbol{U_{a,n}}, \boldsymbol{U_{d,n}}$: The allocation values of arrival and departure traffic for the n-th decision period

$\boldsymbol{U_n\mathcal{D}(U_{a,n}, U_{d,n})}$: traffic allocation decision for the n-th time period.

$\boldsymbol{\Delta_{a,n}}, \boldsymbol{\Delta_{d,n}}$: The incremental delay cost of arrivals and departures within the n-th time period.

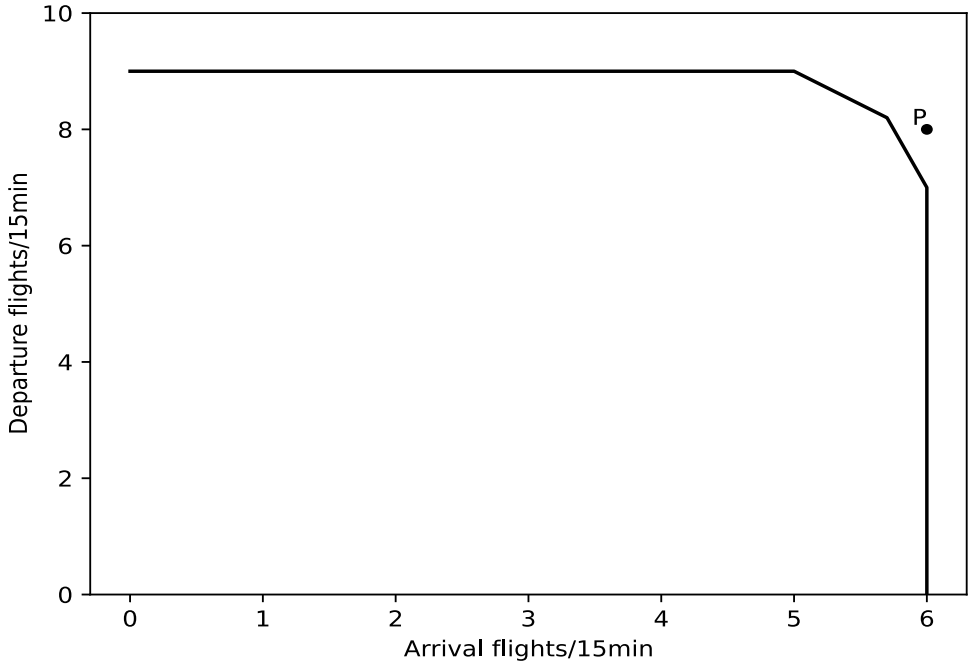



**FIGURE 1.** Airport capacity envelope.

State transition equation:

$$\begin{bmatrix} a_n \\ b_n \end{bmatrix} + \begin{bmatrix} U_{a,n} \\ U_{d,n} \end{bmatrix} = \begin{bmatrix} a_{n-1} \\ d_{n-1} \end{bmatrix} + \begin{bmatrix} \hat{a}_n \\ \hat{b}_n \end{bmatrix} \tag{III.17}$$

Objective function: Minimizes the sum of incremental delay costs for each period

$$min \sum_{n \in T_n} [\Delta_{a,n} + \Delta_{d,n}] \tag{III.18}$$

Using $\Delta_{a,n}$ as an example, let's explain the calculation method for the increment of arrival delay cost:determine whether there exists j $\in \{1, 2, \ldots, n-1\}$,such that $\hat{a}_j + \hat{a}_{j+1} + \ldots + \hat{a}_n \leq a_n \leq \hat{a}_{j-1} + \hat{a}_j + \ldots +, \hat{a}_n$.This condition indicates that the flights delayed until the nth time period have already experienced delays in multiple preceding time periods. If present, then:

$$\begin{aligned} \Delta_{a,n} = (a_n - \sum_{k=j}^{n} \hat{a}_k)(((n-j+2)\Delta T)^{1+\delta} \\ - ((n-j+1)\Delta T)^{1+\delta}) \\ + sum_{k=j}^{n} \hat{a}_k(((n-k+1)\Delta T)^{1+\delta} \\ - ((n-k)\Delta T)^{1+\delta}) \end{aligned} \tag{III.19}$$

If absent,then:

$$\Delta_{a,n} = a_n \Delta T^{1+\delta} \tag{III.20}$$

The constraint conditions:

$$\begin{aligned} \alpha_{i,n}(a_{n-1} + \hat{a}_n - a_n) + \beta_{i,n}(d_{n-1} + \hat{d}_n - d_n) \leq \gamma_{i,n}, \\ \forall n \in T_n, n \neq N+1, i \in I_n \end{aligned} \tag{III.21}$$

$$a_{n-1} + \hat{a}_n \geq a_n, \quad \forall n \in T_n \tag{III.22}$$

$$d_{n-1} + \hat{d}_n \geq d_n, \quad \forall n \in T_n \tag{III.23}$$

$$\Delta_{a,n}, \Delta_{d,n}, \hat{a}_n, \hat{d}_n, a_n, d_n \in N, \quad \forall n \in T_n \tag{III.24}$$

Equation (III.13) represents flights delayed to the nth time period with delays exceeding one time period. Flights delayed to the nth time period and spanning multiple time periods incur an incremental delay cost, calculated as the total delay cost for the flight in the nth time period minus the total delay cost in the (n-1)th time period. $\sum_{k=j}^{n} \hat{a}_k$ represents the sum of arrival demand from the jth time period to the nth time period, where $j$ is the earliest time period affected by delays. $\Delta T^{1+\delta}$ represents the $(1+\delta)$ power of the time interval between time periods where $\delta$ is the coefficient of the super-linear delay cost function. Equation (III.14) describes flights delayed to the nth time period for only one time period, with the incremental delay cost being the delay cost for a single time period delay of that flight. Equation (III.15) indicates that the allocated traffic should meet the capacity envelope requirement within its segmented convex function. Equations (III.16) and (III.17) express that the allocated traffic for each time period should not be negative.

### 2) ALGORITHM DESIGN

The paper define the incremental delay cost based on the current stage's state. Therefore, a network diagram can be constructed as shown in Figure 2. The network consists of $N+2$ stages, each with several states $S_n^{(k)}$. Adjacent stages are connected by paths with fixed lengths, where the

path is defined as the incremental delay cost $\Delta_N^{(k)}$ based on the previous stage's state. Stages 0 and $N+1$ are virtual stages, where all the flow in Stage 0 represents delayed flights, and Stage $N+1$ has an unlimited capacity, allowing all the delayed flights from the previous stage to arrive or depart. Thus, the paper transform the problem of minimizing delay costs into a shortest path problem between network nodes, where the objective is to choose a state at each stage that minimizes the total path length formed by connecting adjacent states. This shortest path problem in the network satisfies the principle of optimality, meaning that the optimal strategy for the entire process must include the optimal strategies for each sub-process.

The shortest path problem in this network satisfies the ''principle of optimality'', which means that the optimal strategy for the entire process must include the optimal strategies for each sub-process. If we separate the entire process from a state $S_n^{(k)}$ at stage $n$ of the optimal strategy, then the optimal strategy for the subsequent sub-process must be contained within the optimal strategy for the entire process. Using dynamic programming to solve the shortest path problem in this network, the paper define the sub-problem's optimal value function $f_n(S_n^{(k)})$. This function represents the shortest path value from the start to the preceding sub-process state $S_n^{(k)}$ when the current stage n is given.

$$f_n(S_n^{(k)}) = \min_{U_1,U_2,\ldots,U_n} \sum_{(t=1)}^{n} r_t(S_t, U_t) \tag{III.25}$$

$$r_t(Z_t, U_t) = \Delta_{a,t} + \Delta_{d,t} \tag{III.26}$$

The recursive relation for the optimal value function of adjacent stage sub-processes is defined as follows:

$$f_n(S_n^{(k)}) = \min_{v \in S_{n-1}} [\Delta_{n-1}^{(v)} + f_{n-1}(S_{n-1}^{(v)})] \tag{III.27}$$

Therefore, the original objective function is transformed into:

$$f_{N+1}(S_{N+1}) = f_N(S_N) = \min \sum_{t \in T_n} [\Delta_{a,t} + \Delta_{d,t}] \tag{III.28}$$

Stage state $S_n$ is a set containing all possible states $(a_n, d_n)$,Among them:

$$a_n \in [0, \sum_{k=1}^{n} \hat{a}_k], d_n \in [0, \sum_{k=1}^{n} \hat{d}_k] \tag{III.29}$$

The minimum number of delayed arrival and departure flights in each time period is 0 (indicating that the demand can be met), while the maximum value is the cumulative flow total demand from previous stages (with a capacity of 0).

The airport capacity optimization problem essentially involves assigning time slots for arrivals and departures of each flight while satisfying the system capacity constraints. The actual decision variables are determined by the arrival and departure situations of flights within each time slot. Taking the allocation of arrival flight flows as an example, after knowing the arrival demand $\hat{a}_t$ each time period and calculating $U_{a,t}$ for each time period, the specific method for specifying the arrival time for flights is as follows:

1) Establish the initial demand list InitList and the delayed flight list DelayList, and initialize them as empty. Initialize n = 1.

2) Sort the inbound flow for time period t in ascending order based on the estimated arrival time and store it in InitList. If $U_{a,n}$,n is less than or equal to the size of DelayList, then assign the first $U_{a,n}$ flights in DelayList to the time period t for arrival and remove these flights from DelayList. Merge all flights in InitList in sequence to the end of DelayList and clear InitList. If $U_{a,n}$ is greater than the size of DelayList, then assign all flights in DelayList and the first $U_{a,n}$ - Size(DelayList) flights in InitList to the nth time period for arrival. Clear DelayList and store the remaining $\hat{a}_n$ - $U_{a,n}$ + Size(DelayList) flights from InitList in sequence to DelayList. Finally, clear InitList.

3) If n is not equal to N, increment n by 1 and return to step 2; otherwise, end the process.

**Algorithm 1** Flight Assignment Algorithm

```
Initialize: DelayList ← ∅, InitList ← ∅, n ← 1
for t = 1 to N do
    Sort: Populate InitList with a^t flights sorted by earliest estimated arrival time
    Assign Flights:
    if U_{a,t} ≤ |DelayList| then
        Assign first U_{a,t} flights from DelayList to time t
        Update DelayList: DelayList ← DelayList[U_{a,t} + 1 :]
    else
        Assign all DelayList flights to time t
        Assign U_{a,t} − |DelayList| earliest flights from InitList to time t
        Update DelayList: DelayList ← InitList[U_{a,t} − |DelayList| + 1 :]
    end if
    Update Lists:
    DelayList ← DelayList ∪ InitList
    InitList ← ∅
end for
Terminate when t = N
```

While our current model provides valuable insights into overall airport capacity, it does not specifically account for the varying impacts of different airport locations such as approaches, individual runways, or terminals. Future research could refine this model by incorporating these location-specific factors, potentially leading to more nuanced and targeted capacity predictions and optimization strategies.

## IV. REAL DATA ANALYSIS

### A. CAPACITY PREDICTION MODEL

This section analyzes a real dataset from a meteorological organization using semi-varying coefficient models to predict

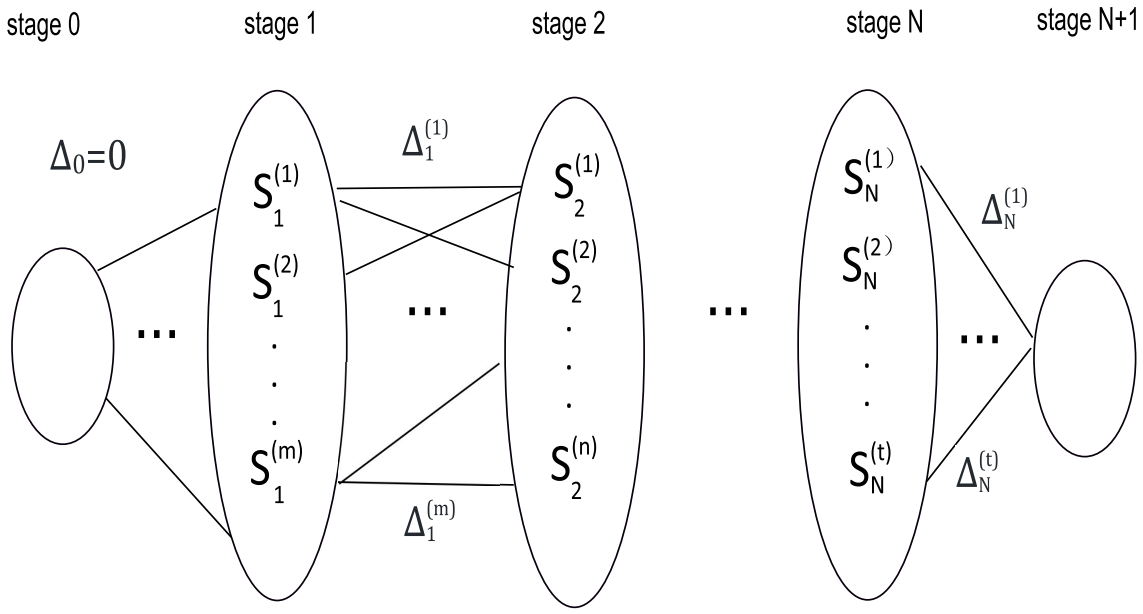



**FIGURE 2.** Network path diagram corresponding to traffic allocation problems.

**TABLE 1.** Multiple predictor variables.

| variable | type | content |
|---|---|---|
| cloud ceiling | continuous | 0-1500 (m) |
| temperature | continuous | 18.1-38.6 (°C ) |
| precipitation | continuous | 0-77.1 (mm) |
| visibility | continuous | 100-30000 (m) |
| wind direction | continuous | 1-360 (°) |
| wind speed | continuous | 0-7.2 (m/s) |
| max wind | continuous | 0.8-17.2 (m/s) |
| weather phenomena | discrete | 0,5,10,20,21,27,28,45,46,50,60,62,64 |
| dew point | continuous | 15.7-29.2 (°C ) |
| corrected pressure | continuous | 995.3-1016.4 (hPa) |
| wind force | discrete | 0,1,2,3,4 |
| thunder | discrete | 0,1 |
| runway1 | discrete | 2,3,4,5,6 |
| runway2 | discrete | 2,3,4,5 |
| AAR | continuous | 2-46 |
| ADR | continuous | 1-46 |

**TABLE 2.** $R^2$ values for the relationships between AAR and various factors.

| Variables | $R^2$ Value | Relationship Type |
|---|---|---|
| Visibility vs. AAR | 0.85 | Linear |
| Wind Speed vs. AAR | 0.80 | Linear |
| Wind Force vs. AAR | 0.78 | Linear |
| Cloud Ceiling vs. AAR | 0.82 | Linear |
| Wind Direction vs. AAR | 0.65 | Nonlinear |
| Maximum Wind Speed vs. AAR | 0.60 | Nonlinear |
| Temp.vs. AAR | 0.58 | Nonlinear |
| Temp. vs. Wind Direction | 0.67 | Nonlinear |
| Temp. vs. Maximum Wind Speed | 0.70 | Nonlinear |

*Remark: The $R^2$ values represent the goodness of fit for each relationship.*

hourly Airport Arrival Rate (AAR) and Airport Departure Rate (ADR) at Guangzhou Baiyun International Airport from May 1 to August 31, 2019. The predictor variables used in this study include cloud ceiling, temperature, precipitation, visibility, wind direction, wind speed, maximum wind speed, weather phenomena, dew point, corrected pressure, wind force, thunder, and number of runways. Notably, discrete variables such as weather phenomena and thunder were preprocessed into numerical representations (through ordinal encoding) to ensure compatibility with correlation analysis and subsequent modeling. For weather phenomena, higher values indicate more severe weather conditions. Similarly, thunder was also encoded numerically to reflect its occurrence, with higher values representing more frequent or intense thunder events. While aircraft type can be a factor in airport operations, this study focuses on macro-level capacity prediction where aggregate flow characteristics are more relevant than individual aircraft specifications. A more precise description for predictors can be found in Table 1.

Figure 3 and Figure 4 display the correlation coefficient matrices of variables at a given significance level. The size of each circle in these figures represents the strength of the linear correlation between variables.

Based on the correlation analysis between AAR, ADR, and other covariates, we selected seven variables that significantly affect AAR: wind direction, visibility, maximum wind speed, wind speed, wind force, cloud ceiling, and temperature. The selection of these variables was based on a correlation threshold of $|r| \geq 0.3$, where only variables with a correlation coefficient greater than or equal to 0.3 were considered significant for inclusion. Similarly, for ADR, we identified seven influential variables: wind direction, visibility, maximum wind speed, wind speed, wind force, temperature, and thunder. The selection for ADR followed the same threshold of $|r| \geq 0.3$, ensuring consistency and reliability in the feature selection process.

Figures 5 and 6 reveal the scatter plot and fitted regression relationship between AAR and 7 covariates. The $R^2$ values

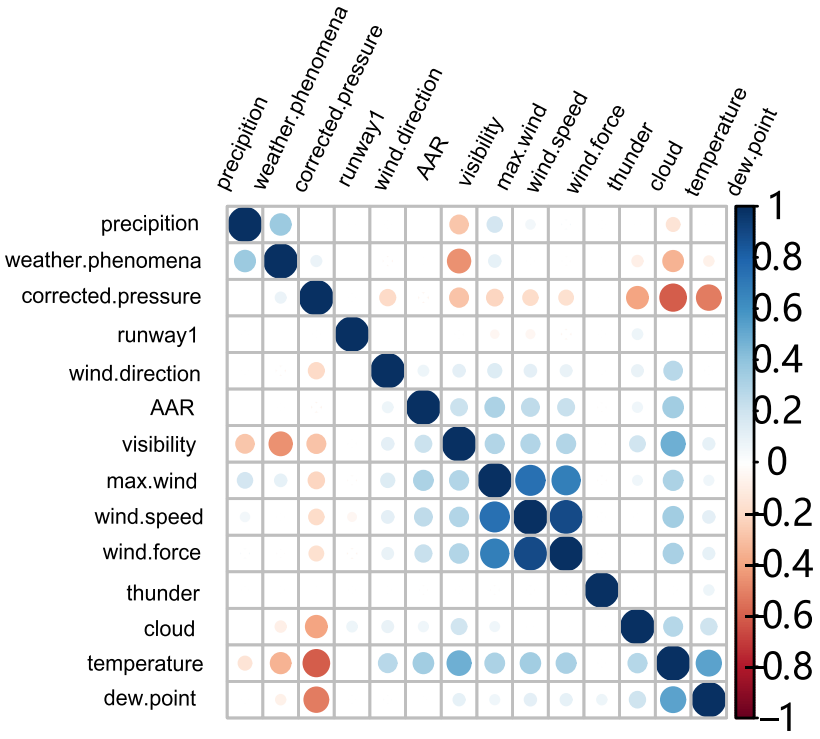



**FIGURE 3.** Matrix diagram of correlation coefficients between AAR and other variables.

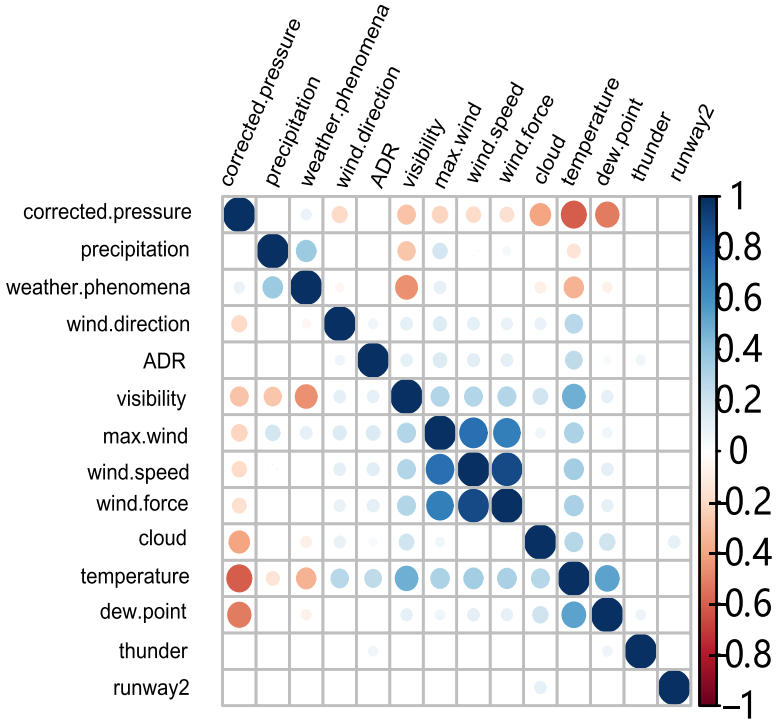



**FIGURE 4.** Matrix diagram of correlation coefficients between ADR and other variables.

for these relationships are summarized in Table 2, which reflects the goodness of fit for each factor affecting AAR. While many flights operate under Instrument Flight Rules (IFR), visibility remains a significant factor affecting overall airport capacity, particularly for ground operations and mixed IFR/VFR traffic.

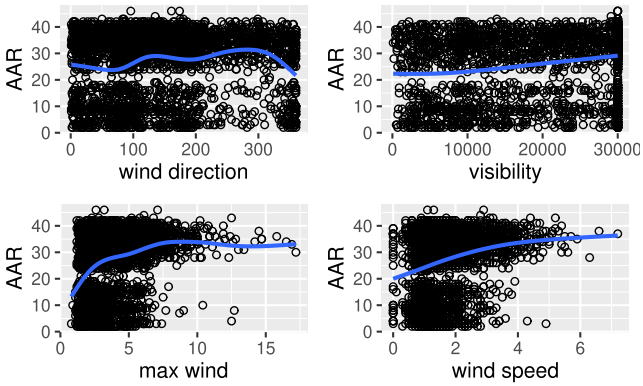



**FIGURE 5.** Scatter plot and interaction between AAR and wind direction(°), visibility(m), max wind(m/s), wind speed(m/s).

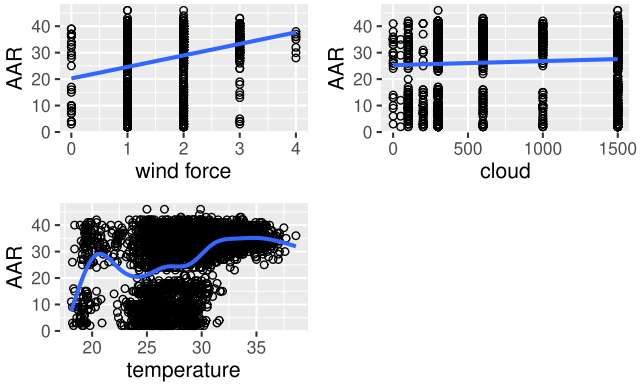



**FIGURE 6.** Scatter plot and interaction between AAR and wind force(wind scale), cloud(m), temperature(°C).

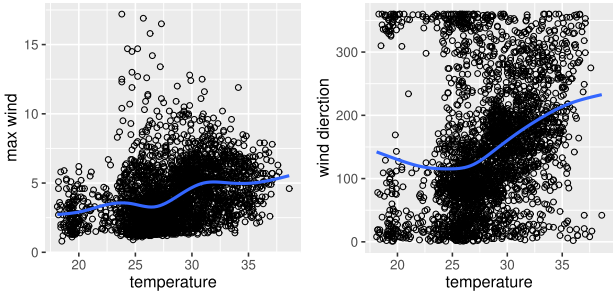



**FIGURE 7.** Scatter plot and interaction between temperature(°C) and max wind(m/s), wind direction(°).

Generally, improved visibility enhances visual conditions for both aircraft and air traffic control systems, allowing pilots to clearly observe runways and navigation markers, reducing risks and potential delays during landing operations. Increases in wind direction variability, maximum wind speed, and overall wind speed can affect aircraft speed and stability, potentially necessitating additional safety measures or reducing airport operational efficiency. Higher cloud cover can reduce pilots' visibility of runways and airport markers, potentially increasing landing risks and consequently affecting the AAR. High temperatures result in lower air density, which can affect aircraft performance during takeoff and landing. This may necessitate longer landing distances or slower descent speeds, impacting the AAR. Moreover, high temperatures can degrade the performance of airport equipment and materials, further impacting operational capabilities and efficiency. From the scatter plots and fitted regression relationships in Figures 5 and 6, it can be seen that, approximately linear relationships between AAR and visibility, wind speed, wind force, and cloud ceiling. Additionally, these figures demonstrate nonlinear relationships between AAR and wind direction, maximum wind speed, and temperature.

Figure 7 illustrates nonlinear relationships between temperature and both wind direction and maximum wind speed. This suggests that temperature acts as a confounding variable when considering the combined effects of wind direction and maximum wind speed on AAR.

Figures 8 and 9 demonstrate the scatter plot and fitted regression relationship between ADR and 7 covariates.

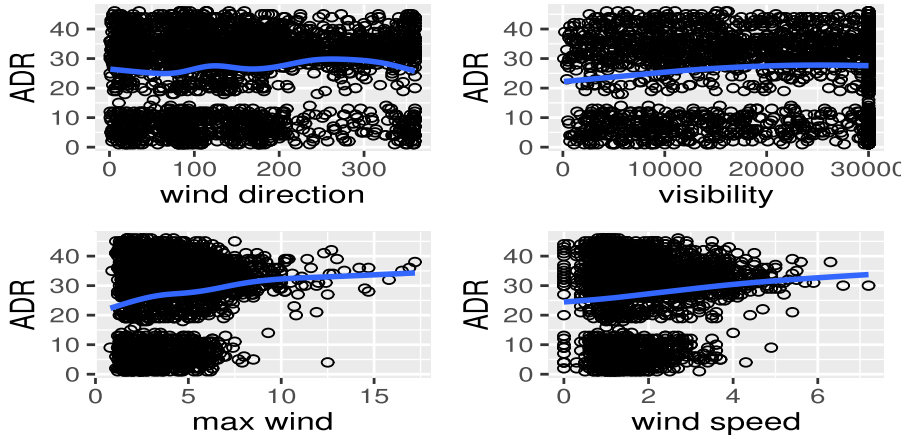



**FIGURE 8.** Scatter plot and interaction between ADR and wind direction(°), visibility(m), max wind(m/s), wind speed(m/s).

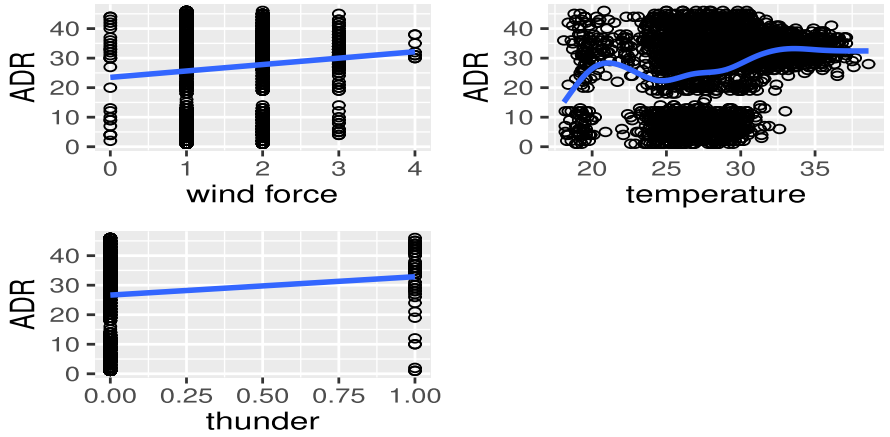



**FIGURE 9.** Scatter plot and interaction between ADR and wind force(wind scale), temperature(°C), thunder(Thunder level).

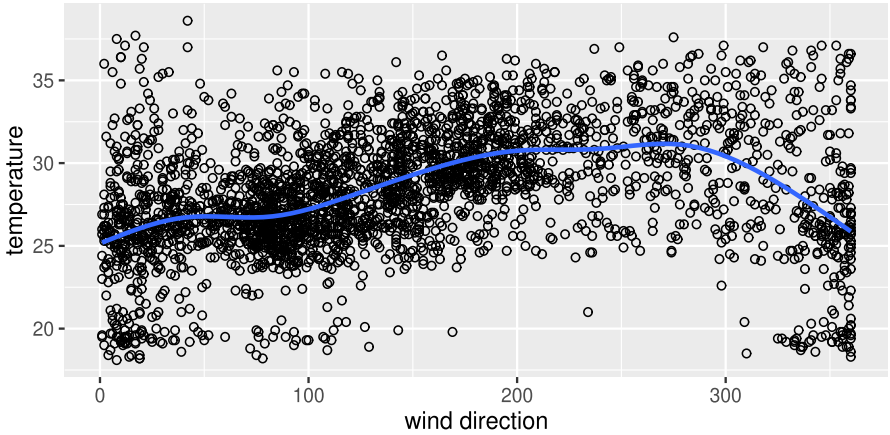



**FIGURE 10.** Scatter plot and interaction between wind direction(°) and temperature(°C).

The $R^2$ values for these relationships are summarized in Table 3, which reflects the goodness of fit for each factor affecting ADR. Good visibility provides pilots with better visual conditions, facilitating the normal execution of takeoff

**TABLE 3.** $R^2$ values for the relationships between ADR and various factors.

| Variables | $R^2$ Value | Relationship Type |
|---|---|---|
| Visibility vs. ADR | 0.83 | Linear |
| Maximum Wind Speed vs. ADR | 0.85 | Linear |
| Wind Speed vs. ADR | 0.88 | Linear |
| Wind Force vs. ADR | 0.80 | Linear |
| Thunder Occurrence vs. ADR | 0.87 | Linear |
| Wind Direction vs. ADR | 0.68 | Nonlinear |
| Temp. vs. ADR | 0.62 | Nonlinear |
| Temp. vs. Wind Direction | 0.66 | Nonlinear |

*Remark: The $R^2$ values represent the goodness of fit for each relationship.*

**TABLE 4.** Data structure of AAR and ADR.

| variable | notation | variable | notation |
|---|---|---|---|
| cloud ceiling | $X_{A1}$ | visibility | $X_{D1}$ |
| visibility | $X_{A2}$ | max wind | $X_{D2}$ |
| wind speed | $X_{A3}$ | wind speed | $X_{D3}$ |
| wind force | $X_{A4}$ | wind force | $X_{D4}$ |
| wind direction | $Z_{A1}$ | thunder | $X_{D5}$ |
| max wind | $Z_{A2}$ | temperature | $Z_{D1}$ |
| temperature | $U_A$ | wind direction | $U_D$ |
| AAR | $Y_A$ | ADR | $Y_D$ |

procedures and consequently improving the Airport Departure Rate (ADR). Higher wind force, variable wind direction, and increased wind speed can directly impact the safety and performance of takeoff operations, potentially limiting aircraft takeoff capabilities. The presence of thunder and extreme temperatures represent adverse weather conditions that can have direct and unfavorable impacts on takeoff operations, thus affecting ADR. From the scatter plots and fitted regression relationships in Figures 8 and 9, it can be seen that, approximately linear relationships between ADR and visibility, maximum wind speed, wind speed, wind force, and thunder occurrence, nonlinear relationship with wind direction and temperature.

Figure 10 demonstrates a nonlinear relationship between wind direction and temperature. Consequently, wind direction is considered a confounding variable when assessing the combined effect of temperature on ADR. Based on the above analysis, Table 4 summarizes the variables used in the AAR and ADR prediction models.

Using the variables identified in Table 4, we can express the full models for AAR and ADR as follows:

$$\begin{aligned}\text{Model.I}\quad \boldsymbol{Y}_A =&\beta_{A1}\boldsymbol{X}_{A1}+\beta_{A2}\boldsymbol{X}_{A2}+\beta_{A3}\boldsymbol{X}_{A3}+\beta_{A4}\boldsymbol{X}_{A4}\\&+\alpha_1(U_A)\boldsymbol{Z}_{A1}+\alpha_2(U_A)\boldsymbol{Z}_{A2}+\boldsymbol{\varepsilon}_A,\\ \text{Model.II}\quad \boldsymbol{Y}_D =&\beta_{D1}\boldsymbol{X}_{D1}+\beta_{D2}\boldsymbol{X}_{D2}+\beta_{D3}\boldsymbol{X}_{D3}+\beta_{D4}\boldsymbol{X}_{D4}\\&+\beta_{D5}\boldsymbol{X}_{D5}+\alpha_1(U_D)\boldsymbol{Z}_{D1}+\boldsymbol{\varepsilon}_D,\end{aligned}$$

The predicted value of AAR and ADR is:

$$\begin{aligned}\text{Model.I}\quad \boldsymbol{Y}_A =& 0.0337\boldsymbol{X}_{A1}+3.5617\boldsymbol{X}_{A2}-1.3688\boldsymbol{X}_{A3}\\&-1.0765\boldsymbol{X}_{A4}+\alpha_1(U_A)\boldsymbol{Z}_{A1}+\alpha_2(U_A)\boldsymbol{Z}_{A2}+\boldsymbol{\varepsilon}_A,\\ \text{Model.II}\quad \boldsymbol{Y}_D =&-1.4494\boldsymbol{X}_{D1}+12.7389\boldsymbol{X}_{D2}-2.9240\boldsymbol{X}_{D3}\\&+0.0311\boldsymbol{X}_{D4}+5.9512\boldsymbol{X}_{D5}+\alpha_1(U_D)\boldsymbol{Z}_{D1}+\boldsymbol{\varepsilon}_D,\end{aligned}$$

**TABLE 5.** The point estimate results of $\beta$.

| variables | estimates | std.error |
|---|---|---|
| $\hat{\beta}_{A1}$ | 0.0337 | 0.7744 |
| $\hat{\beta}_{A2}$ | 3.5617 | 3.9832 |
| $\hat{\beta}_{A3}$ | -1.3688 | 0.7345 |
| $\hat{\beta}_{A4}$ | -1.0765 | 0.5875 |
| $\hat{\beta}_{D1}$ | -1.4494 | 0.8063 |
| $\hat{\beta}_{D2}$ | 12.7389 | 2.5734 |
| $\hat{\beta}_{D3}$ | -2.9240 | 4.2356 |
| $\hat{\beta}_{D4}$ | 0.0311 | 0.7837 |
| $\hat{\beta}_{D5}$ | 5.9512 | 1.6440 |

*Remark: The estimates stand for the estimates of coefficients, and std.error represents standard error of those estimators.*

The parameter estimation and the fitting curve of varying-coefficient function can be obtained by R program, see Table 5, Figure 11 and Figure 12. To further compare the semi-varying coefficient model and varying-coefficient model with AAR and ADR, the paper use 5-fold cross-validation (CV) to compare which model has better out-of-sample prediction ability. Cross-validation is a widely used technique in data mining and machine learning for assessing how well selected models and parameters generalize to independent, out-of-sample datasets. This technique is particularly valuable when the primary goal is prediction on new, unseen data rather than fitting the model to the existing sample. Table 6 presents the results of the cross-validation analysis.

The definition of the $R^2_{CV}$ is that,

$$\begin{aligned}R^2_{CV} &= \frac{1}{5}\sum_{j=1}^{5}=\left(1-\frac{SSE_j}{SST_j}\right)\\&=\frac{1}{5}\sum_{j=1}^{5}\left(1-\frac{\sum_{i=1}^{n_j}(\hat{y}_{j,i}-y_{j,i})^2}{\sum_{i=1}^{n_j}(y_{i,j}-\bar{y}_j)^2}\right).\end{aligned}$$

By the above formula, the $R^2_{CV}$ being larger means $SSE_j$ becoming smaller. And each $SSE_j$ means the out-of-sample sum squares of errors, hence a smaller $SSE_j$ indicates a better prediction of the model to the out-of-sample data. From Table 6 we can see that, the change of $Model.I$ and $Model.II$ reduces less than $Model.I(varing-coefficient)$ and $Model.II(varing-coefficient)$ [13]. The less reduction of $R^2$ means more reliable prediction of the models. In summary, semi-varying coefficient model has been shown to be more attractive.

### B. FLOW ALLOCATION OPTIMIZATION MODEL

We utilized flight data for arrivals and departures at Guangzhou Baiyun Airport from 7:00 AM to 9:00 AM on August 31, 2019, to optimize and coordinate flow allocation. Using the capacity prediction model described in Section III-A, we generated the flight capacity curve depicted in Figure 13. We divided this optimization period into eight 15-minute time slots, making traffic allocation decisions for

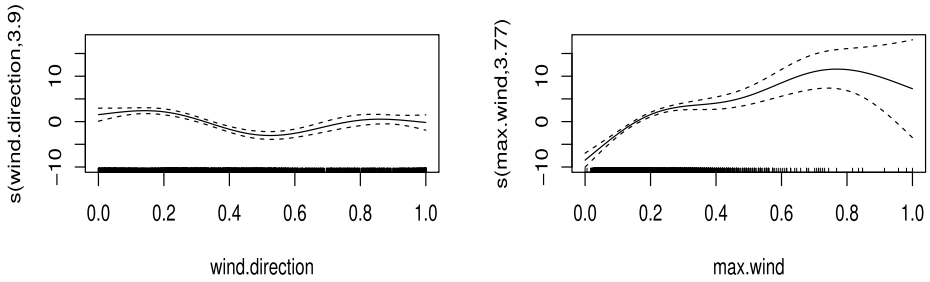



**FIGURE 11.** The fitted curves and asymptotic normal interval estimates with 95% confidence of $\hat{\alpha}_1(U_A)$ and $\hat{\alpha}_2(U_A)$.

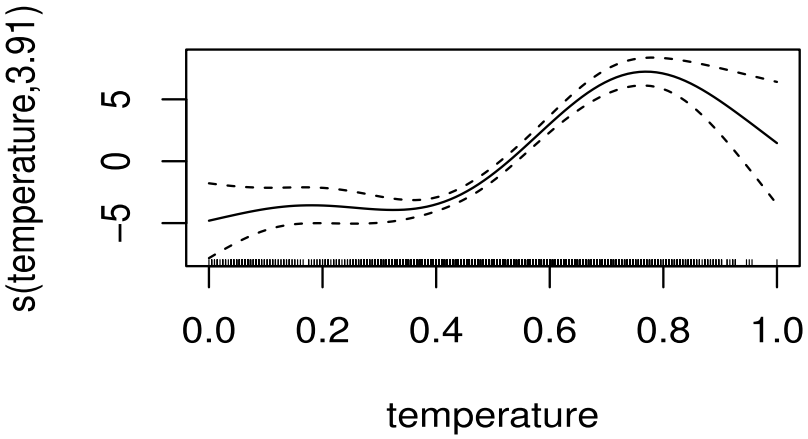



**FIGURE 12.** The fitted curves and asymptotic normal interval estimates with 95% confidence of $\hat{\alpha}_1(U_D)$.

**TABLE 6.** Results of airport rate prediction model cross-validation.

| Model | Original $R^2$ | $R^2_{CV}$ | Change |
|---|---|---|---|
| *Model.I* | 0.214 | 0.1662 | 0.048 |
| *Model.I*(*varing* − *coefficient*) | 0.224 | 0.1674 | 0.057 |
| *Model.II* | 0.102 | 0.0710 | 0.031 |
| *Model.II*(*varing* − *coefficient*) | 0.103 | 0.0675 | 0.035 |

*Remark:*$R^2$ *is a statistic that will give some information about the goodness of fit of a model, Original* $R^2$ *represents the original* $R^2$ *of the model,* $R^2_{CV}$ *represents the R-Square of model after doing the 5 fold cross validation , Change means the result of original* $R^2$ *minus* $R^2_{CV}$.

each slot. Table 7 presents the estimated arrival and departure traffic demand for each time slot, along with the optimized traffic allocation strategy and the resulting number of delayed flights. We assumed maximum flight capacities of 7 for arrivals and 8 for departures at the airport.

Table 8 presents a detailed breakdown of the traffic distribution for the optimization time slot from 8:45 to 9:00. Analysis of Table 7 reveals that, due to the absence of departure delays in the preceding time slot (8:30 to 8:45), all allocated departure flights for the current time slot (8:45 to 9:00) correspond to the original departure demand. With a departure demand of 8 flights and an allocation of only 7 departures, one flight from the original departure demand will be delayed. For arrivals, the model prioritizes the 3 delayed flights from the previous time slot (8:30 to 8:45), followed by the 2 earliest estimated arrivals from the current slot, totaling the 5 allocated arrival flights. Given the arrival demand of 7 flights and the allocation of 5 arrivals, 5 flights from the original arrival demand for this time slot will be delayed.

To accurately represent the interdependence between arrival and departure capacities, we express the airport capacity as an envelope curve encompassing both arrival and departure capacities. Figure 14 illustrates the comparison between arrival and departure flows and capacities for each time period, both before and after optimization. Black points represent the pre-optimization state, while red points depict the post-optimization scenario. Figure 14 demonstrates that the optimized traffic allocation falls within the airport's capacity envelope, confirming that the proposed allocation strategy satisfies the airport's capacity constraints.

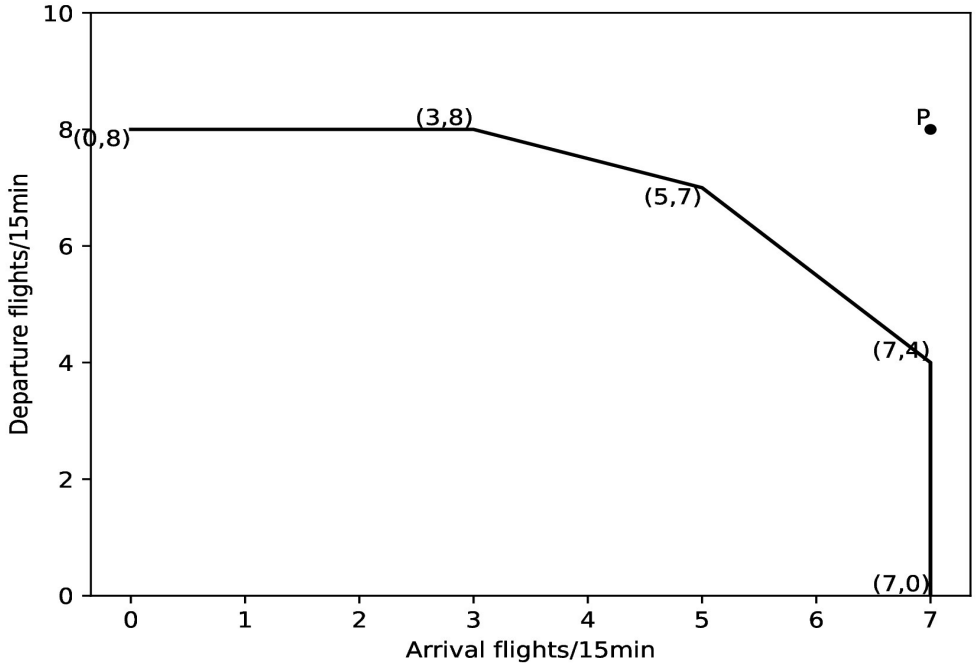



**FIGURE 13.** The flight capacity curve.

**TABLE 7.** The flow allocation strategy.

| Optimization period | Arrival demands | Departure demands | Arrival distribution | Departure distribution | The number of delayed arrivals | The number of delayed departures |
|---|---|---|---|---|---|---|
| 7:00-7:15 | 3 | 8 | 3 | 8 | 0 | 0 |
| 7:15-7:30 | 5 | 4 | 4 | 3 | 1 | 1 |
| 7:30-7:45 | 6 | 4 | 6 | 5 | 1 | 0 |
| 7:45-8:00 | 4 | 7 | 5 | 7 | 0 | 0 |
| 8:00-8:15 | 5 | 8 | 5 | 7 | 0 | 1 |
| 8:15-8:30 | 6 | 4 | 6 | 5 | 0 | 0 |
| 8:30-8:45 | 6 | 8 | 3 | 8 | 3 | 0 |
| 8:45-9:00 | 7 | 8 | 5 | 7 | 5 | 1 |

**TABLE 8.** The traffic distribution during the optimized time of 8:45-9:00.

| Arrival or departure flight | Delayed | The expected arrival/ departure time | Optimized arrival/ departure time |
|---|---|---|---|
| $A1$ | true | 08:30 | 8:45-9:00 |
| $A2$ | true | 08:32 | 8:45-9:00 |
| $A3$ | true | 08:34 | 8:45-9:00 |
| $A4$ | false | 08:46 | 8:45-9:00 |
| $A5$ | false | 08:55 | 8:45-9:00 |
| $D1$ | false | 08:46 | 8:45-9:00 |
| $D2$ | false | 08:48 | 8:45-9:00 |
| $D3$ | false | 08:50 | 8:45-9:00 |
| $D4$ | false | 08:55 | 8:45-9:00 |
| $D5$ | false | 08:57 | 8:45-9:00 |
| $D6$ | false | 08:58 | 8:45-9:00 |
| $D7$ | false | 08:58 | 8:45-9:00 |

*Note: Flights with a prefix "A" in their flight numbers are arrival flights, while flights with a prefix "D" are departure flights.

To further refine our analysis, it is worth considering how our results might vary if we were to examine different airport locations separately. For instance, our analysis could be refined by considering the following airport-specific factors:

1)Approaches: Various approach paths may exhibit different sensitivities to weather conditions or air traffic control constraints, potentially resulting in location-specific capacity predictions.

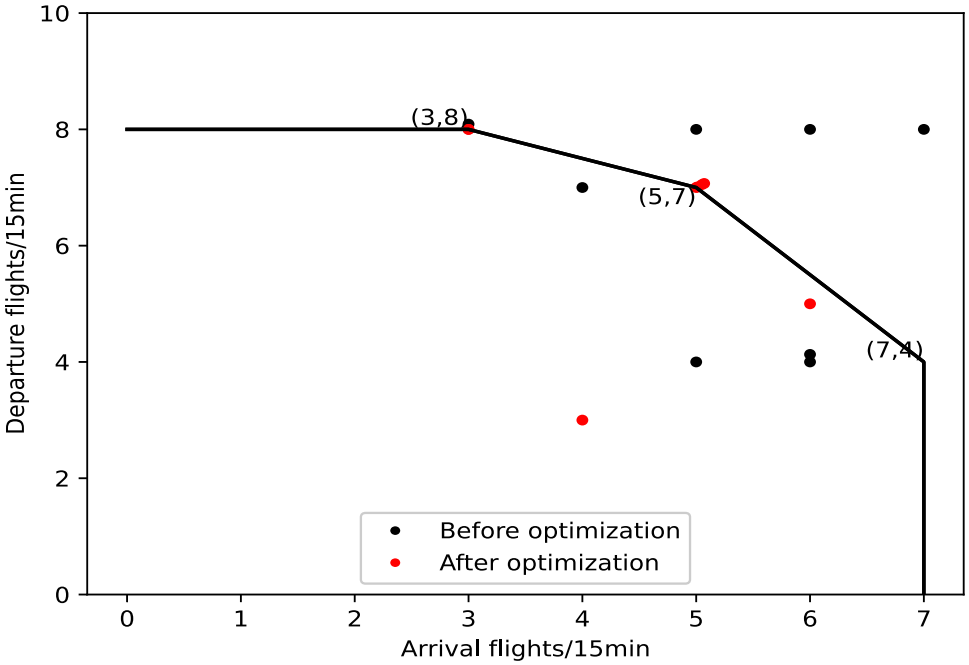



**FIGURE 14.** Matching of capacity flow before and after optimization.

2)Runways: Individual runways often possess unique operational characteristics (e.g., length, instrumentation, orientation) that could influence their capacity under different weather and traffic conditions.

3)Terminals: Terminal capacity may impose additional constraints, especially for departures, which are not accounted for in our current airside-focused model. This could include factors such as gate availability and passenger processing times.

4)Ground movement areas: The layout and efficiency of taxiways and aprons can significantly influence overall airport capacity, particularly during peak times or adverse weather conditions. Factors such as taxiway congestion and de-icing pad locations could be considered.

## V. CONCLUSION AND FUTURE DIRECTIONS

Weather variability introduces uncertainty in future airport capacity, complicating flight time slot allocation and potentially leading to various economic losses. In this study, we developed a capacity prediction model using a semi-varying coefficient approach, based on hourly Airport Arrival Rate (AAR) and Airport Departure Rate (ADR) data from a real meteorological dataset. Subsequently, we developed an optimization model for allocating arrival and departure flows, incorporating airport capacity constraints with the objective of minimizing total delay costs for both arriving and departing flights. Our model aims to determine an optimal air traffic allocation strategy that simultaneously minimizes flight delays, maximizes capacity utilization, and effectively coordinates traffic distribution. Our research findings provide a scientific foundation for decision-makers in air traffic management and airport operations, including airport dispatchers and aviation regulatory agency managers. These results can inform more data-driven decisions and inspire new approaches to tactical and pre-tactical airspace traffic management. To illustrate the practical application of our research, we present specific scenarios for key decision-makers:1. Airport Operations Managers: When facing approaching weather systems, they can use our model to estimate impacts on arrival and departure rates, adjusting resources accordingly.2. Air Traffic Flow Managers: During high-demand periods, they can optimize slot allocation to minimize delays and maximize runway utilization.3. Airline Schedule Planners: Our capacity predictions can inform decisions about flight timing and frequency, especially for seasonal planning.4. Airport Infrastructure Planners: The model can assess potential capacity gains from various improvement projects under different weather conditions.5. Regulatory Bodies: Aviation authorities can use our model to set realistic capacity targets and evaluate air traffic management initiatives. These scenarios demonstrate how our research translates into actionable insights for aviation stakeholders, providing a framework for data-driven decision-making in complex airport environments.

This study constructs an effective airport capacity prediction and flow allocation optimization model based on data from Guangzhou Baiyun Airport. However, we acknowledge limitations in our study, primarily the reliance on a single data source, which may impact the model's generalizability to other airports or regions. Furthermore, future research could employ Monte Carlo simulations to quantify the effects of aircraft mix on capacity predictions, potentially refining our

model with an aircraft mix factor. In the future, we plan to conduct further validation at different types and sizes of airports to enhance the generalizability and practicality of the model. While this study does not incorporate aircraft type data, future research could explore the impact of fleet mix on capacity predictions. However, the current model provides a robust framework for capacity prediction based on key environmental and operational factors that are universally applicable across airports.

Beyond providing optimal strategies for flow allocation, our research has significant managerial implications for airport operations and air traffic management. These implications can help reduce the imbalance between demand and supply in airport capacity utilization. Firstly, our capacity prediction model enables proactive decision-making. Airport managers can anticipate capacity constraints and adjust operations accordingly, potentially implementing demand management strategies such as slot allocation or peak pricing during predicted low-capacity periods. Secondly, the flow allocation optimization model provides a framework for collaborative decision-making between airports and airlines. By sharing predicted capacity and optimal flow allocations, stakeholders can collectively manage demand, potentially through voluntary schedule adjustments or negotiated slot swaps. Thirdly, the insights from our model can inform long-term strategic planning. Persistent capacity shortfalls identified by the model may justify infrastructure investments or airspace redesign initiatives. Lastly, our approach emphasizes the importance of data-driven management in air traffic operations. Airports and air navigation service providers should invest in robust data collection and analysis capabilities to continually refine and apply similar models in their decision-making processes.

Future research could explore the integration of our capacity prediction and flow optimization models with demand management strategies. This could include studying the impact of various pricing mechanisms, slot allocation policies, or collaborative decision-making processes on balancing demand and supply. Furthermore, extending our model to incorporate network effects across multiple airports could yield valuable insights for system-wide air traffic management, particularly in densely connected airspaces. Moreover, future studies could further analyze the specific impact of visibility on IFR vs. VFR operations, leading to a more nuanced treatment of visibility in our capacity prediction model. This could involve a detailed examination of historical weather and flight operation data to better understand the relationship between visibility conditions and actual airport capacity utilization. Future research should also consider the economic implications of implementing such a system, including cost-benefit analyses for airports of different sizes and in various geographical locations. As air traffic continues to grow globally, the importance of efficient capacity utilization and flow management will only increase. Our model provides a foundation for addressing these challenges in a data-driven, systematic manner. Collaboration with industry partners and regulatory bodies will be crucial in further developing and implementing these models in real-world settings.

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